

FULL STACK DATA SCIENCE WITH GENERATIVE AI PROGRAM

Get Future-Ready with **Agentic AI** Specialisation



www.theiscale.com

ABOUT THE PROGRAM

THE FULL STACK DATA SCIENCE WITH GENERATIVE AI PROGRAM PREPARES YOU WITH ESSENTIAL SKILLS IN DATA SCIENCE, MACHINE LEARNING, AND DEEP LEARNING, ALONG WITH HANDS-ON TRAINING IN GENERATIVE AI. THROUGH REAL-WORLD PROJECTS, YOU'LL GAIN PRACTICAL EXPERTISE TO BUILD AI-POWERED SOLUTIONS.

SPECIALITY

THIS PROGRAM FOCUSES ON GENERATIVE AI AND AGENTIC AI ALONG WITH CORE DATA SCIENCE. YOU'LL GAIN HANDS-ON EXPERIENCE WITH TOOLS LIKE LLMS, GPT, AND DIFFUSION MODELS, BLENDING THEORY WITH PRACTICE TO SOLVE REAL-WORLD PROBLEMS AND EXCEL IN TODAY'S AI-DRIVEN WORLD.

AI-POWERED FUTURE

UPSKILL TODAY TO STAY RELEVANT AND LEAD IN THE INNOVATIVE ERA OF ARTIFICIAL INTELLIGENCE.



AI



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Speciality of This Program

Generative AI and Agentic AI are rapidly growing, and our program offers learning opportunities in both.

Generative AI



Generative AI is transforming content creation, coding, marketing, and design. Companies are rapidly adopting AI-powered automation, creating huge career opportunities in tech, research, and business.

Agentic AI

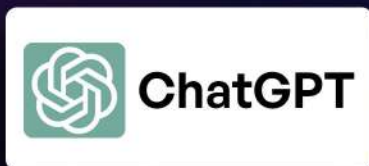


Agentic AI enables systems to plan, make decisions, and execute tasks autonomously, reducing manual work. It is key to developing self-learning, autonomous AI that can power next-gen robotics, virtual assistants, and intelligent business solutions.

The Era of Generative AI

This Gen AI Era Transforms Imagination into Reality, Making Creative AI Tools a Cornerstone of Modern Industry and Life.

Revolutionizing Tomorrow with AI Tools and Technology



Building Industry Skills for Success in the AI-Driven World.

In an AI-driven world, learning data science skills has become essential. Data science is the backbone of artificial intelligence, enabling us to analyze vast datasets, uncover patterns, and make data-informed decisions that drive innovation.

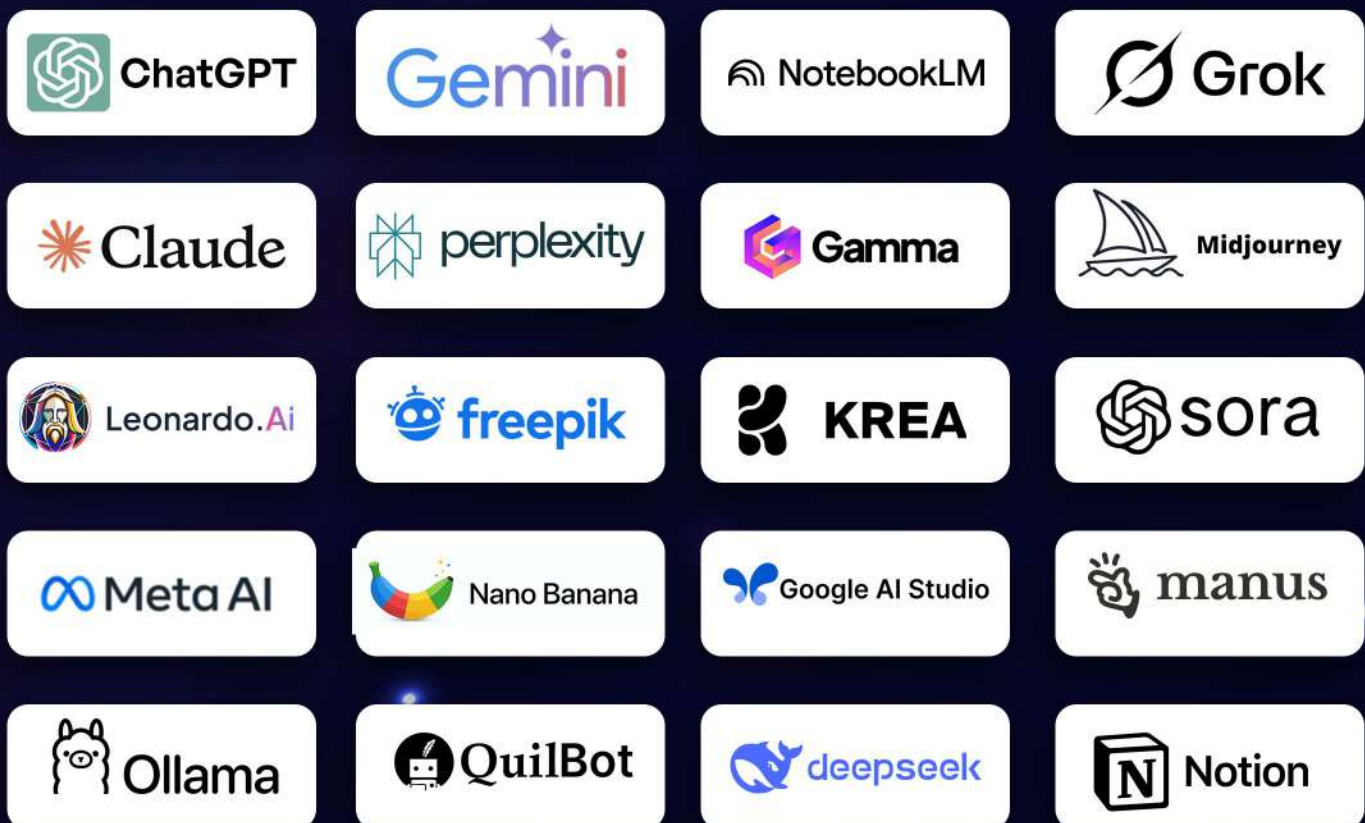


Speciality of This Program

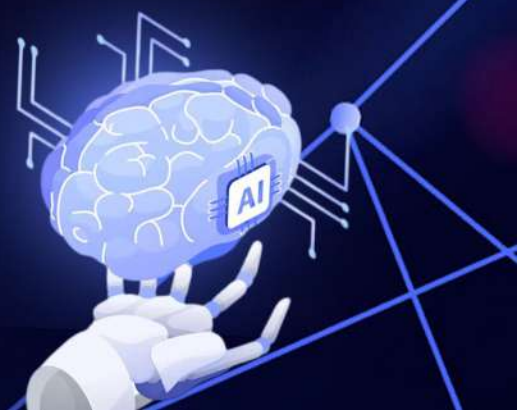
This program takes you on a complete deep dive into AI, from the foundations of Machine Learning, Deep Learning, NLP, and Computer Vision to the latest in Generative AI and Agentic AI. Along with this, you'll also learn to practically use 20+ powerful AI tools across different use cases,

20+ AI Tools

Shape Your Future with 20+ Advanced AI Tools



& Many More



WHY DATA SCIENTIST WITH GEN AI?

Hottest Job of the 21st Century

Data is transforming the world, and becoming data-literate will pay dividends for your career. Data science combined with artificial intelligence, is the fastest growing field and will continue expanding over the next decade. With many companies now relying on data-driven operations, data analysis has become essential. This increasing dependence on data is driving the demand for data scientists skilled in artificial intelligence, a trend that will only grow stronger in the future.



Data Scientist with AI expertise is becoming the number one emerging job role in the world.



The data scientist and AI position is the **highest-paying entry-level job** in the world.



Only 25% of new employees feel confident enough to use organizational data.



54% of the working population need significant reskilling & upskilling.



The global demand for data Scientists and AI professionals is expected to increase by **20 million by 2027.**



\$667.9 billion is Estimated Generative AI market size by 2030, growing at a CAGR of **47.5%**

Diverse Role In Industry



Data Scientist



ML Engineer



AI Engineer



Data Analyst



Deep Learning Engineer



AI Research Scientist



BI Developer



Data Research Scientist

Top Companies for Data Scientist & AI Jobs

Data Scientist and AI jobs are in high demand, and top companies are always looking for skilled talent. This is your chance to learn, grow, and build a strong network in leading firms. Take a look at some of the top companies.



About the Program



Beginner



Intermediate



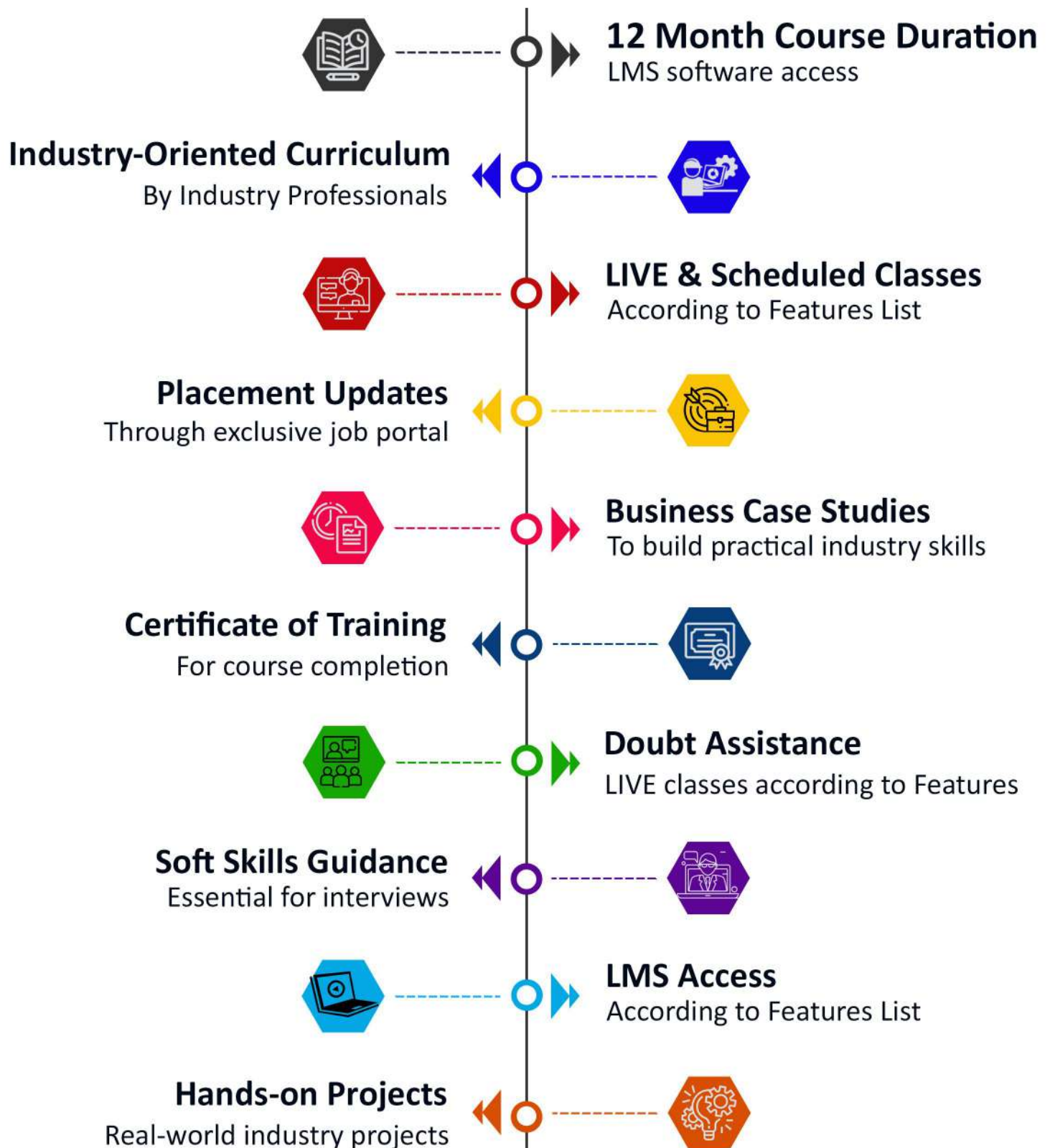
Advanced

Embark on a transformative learning experience with our online program tailored for aspiring data scientists and AI enthusiasts. This comprehensive curriculum is designed to instill proficiency in industry-standard tools, statistical analysis, and data science and AI-driven techniques. Through hands-on projects and expert-led instruction, you'll not only build a strong foundation in data science with generative AI but also gain practical skills to excel in real-world scenarios. With a focus on flexibility and career development support, this program empowers you to launch a successful career in the dynamic and rapidly growing field of Data Science with Generative AI.

Our program prioritizes flexibility, allowing you to shape your learning experience to fit your schedule. It also supports career development, preparing you for success in competitive landscape of Data Science with artificial intelligence.

Program Highlights

Immerse yourself in a curriculum that seamlessly integrates theoretical knowledge with hands-on projects, ensuring a comprehensive skill set in data science and artificial intelligence.



Dedicated Career Support

- According to features list

ATS Resume Building - An excellent resume has the power to unlock countless opportunities. We help you create ATS-friendly resumes that stand out and open doors to success.



Mock Interviews - Mock Interviews help candidates to reduce their stress and help boost their confidence. You will also learn to crack interviews with our interview preparation sessions.

Portfolio - A portfolio is a snapshot of all the projects done and skills required during the program that is shareable across media channels. This will help you to showcase your expertise to potential recruiters.



Exclusive Job Portal – Unlock unparalleled career prospects with our exclusive job portal, facilitating seamless connections with industry-leading companies. Dive into a world of opportunities and fast-track your journey to professional success.

Linkedin Profile Building - Strong LinkedIn profile can contribute to career development. We help you in making a professional LinkedIn Profile.



Program Outcomes

- According to features list

- Acquire proficiency in industry-standard tools such as Python, SQL, and data visualization libraries to manipulate and analyze diverse datasets effectively.
- Apply theoretical knowledge to practical scenarios through hands-on projects, building a robust portfolio that demonstrates your ability to tackle diverse challenges encountered in the field.
- Master the art of conveying insights through compelling data visualizations, using tools like Matplotlib, Seaborn, PowerBI and Tableau to communicate complex findings in a clear and impactful manner.
- Complete the program with a portfolio showcasing practical projects, a comprehensive grasp of data science with generative AI methodologies, and the expertise needed to launch a successful career journey as a Data Scientist and AI professional.



DUAL CERTIFICATE

IN TWO MOST IN-DEMAND AND HIGHLY VALUED SKILLS

**DATA
ANALYTICS**

+

**DATA SCIENCE
WITH GENERATIVE AI**



Transform Your Career with Certificate in Data Science Specializing in Generative AI and Data Analytics. This Industry - Recognized Certification Empowers you with Essential Skills for Today's Data - Driven World.

Get the Certificate of Course Completion

Upon completion of this Course, you will receive a verified, industry - recognized certificate from our training platform.



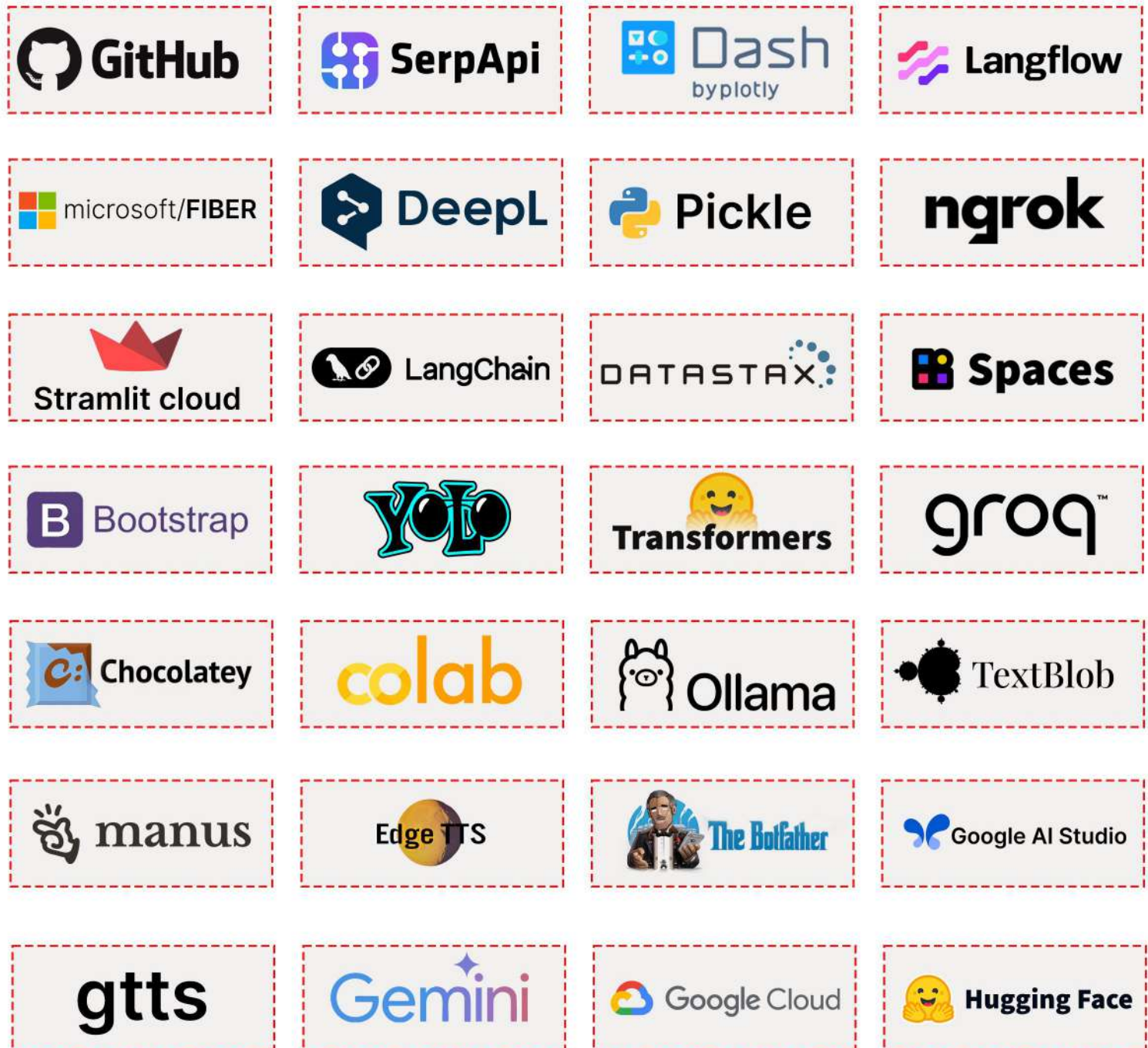
Tools & Frameworks

Learners build real-world portfolio projects that help them master industry-standard tools, frameworks & libraries that are used daily in thousands of companies around the world.



Tools & Frameworks

Practical experience comes from working on real-world projects using industry-standard tools and frameworks. By building projects with these tools,



The Learning Path

Module 1

Fundamentals of Python



Module 2

Python Libraries



Module 3

Statistics



Module 4

Feature Engineering



Module 5

Exploratory Data Analysis

Module 6

Machine Learning - I Supervised



Module 7

Machine Learning - II Unsupervised



Module 8

Deep Learning

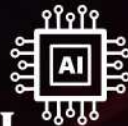


Module 9

Natural Language Processing - NLP



Module 10 
Computer Vision (Open CV)

Module 11 
Generative AI

Module 12 
Agentic AI

Module 13 
20+ AI Tools

Module 14 
SQL

Module 15 
Power BI

Module 16 
Advanced Excel

Module 17 
Tableau

Module 18
Placement Preparation Module

Hands on Industry Oriented Projects



**Price Comparison
Tool**



**RAG
AI Agent**



**Movie Recommendation
Project**



**Covid 19 Analysis
Using Dash Library**



**Whatsapp Chat
Analysis**



**Weather
Prediction**



**Diabetes
Prediction**



**Medical
Prediction**



**Diamond Price
Prediction**

Hands on Industry Oriented Projects

10 PROJECT

Credit Card
Fault Detection

11 PROJECT

Object
Racking

12 PROJECT

Image
Classification

13 PROJECT

Image to
Text

14 PROJECT

Indian Premier
League

15 PROJECT

Shipping
E-Commerce

16 PROJECT

Virat Kohli ODI T20
Performance Index

17 PROJECT

Netflix Stock
Prediction

18 PROJECT

UNICEF Global
Data Analysis

Hands on Industry Oriented Projects

19 PROJECT

Language Detection

Input: こんにちは、世界!

Language Detection

Japanese

Language
Detection

20 PROJECT



Exploratory Data
Analysis

21 PROJECT

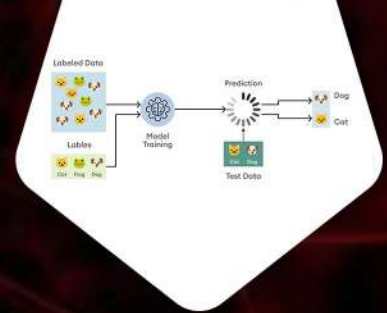


Image
Classification

22 PROJECT



Speech to
Text

23 PROJECT



Spam
Detection

24 PROJECT



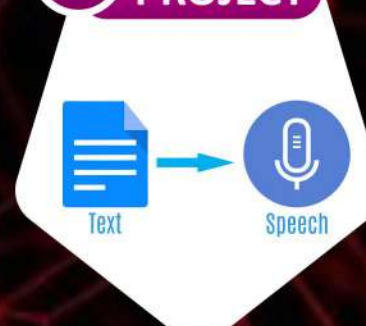
Myntra Sales
Dashboard

25 PROJECT



Rating
Predictions

26 PROJECT



Text to
Speech

27 PROJECT



AI
Agent

Hands on Industry Oriented Projects

28 PROJECT



Streaming Platform Analysis

29 PROJECT



Telegram Chatbot with LLM

30 PROJECT



Music Store Analysis

31 PROJECT



Plant Disease Agent

32 PROJECT



IMDb Web Scrapping with Flask

33 PROJECT



Google Image Scrapping with Flask

34 PROJECT



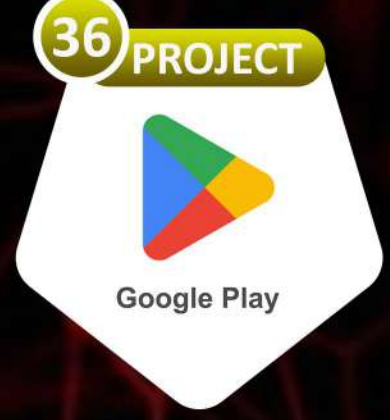
Automobile Bike Analysis

35 PROJECT



Student Performance Indicator

36 PROJECT



Google Playstore Analysis

Academic Team

For helping you in your learning Journey

Nishant Dhote

- Founder & Serial Entrepreneur
- Ex Amazon - Data Science Consultant
- Industry Exp – 10+ Years

THE  iSCALE



Swati

- CBO – The iSCALE
- Ex Byjus – Product Analyst
- Social Projects (Ministry of Youth Affairs)

 BYJU'S



Vibhor Sahu

- IIM - Raipur
- Associate Manager -Tredence Inc.
- Ex-Companies – TCS & Pixel

 **Tredence**
Connect the Dots



Bhavesh

- IIT - Dhanbad
- Placement Head
- Industry Exp – 8+ Years



Bijay

- Berlin School of Business Germany
- Marketing Manager –Allmyhomes
- Ex-Companies Sygns, Internetwarriors, Felmo

 ALLMYHOMES



Sweta Agrawal

- 5Years+ Work Experience
- Data Scientist
- Projects with Int. Clients – Zigtal, Vintage Wine Estates

 BenchPrep



Connected CEOs & CHROs

Prasad Menon
CHRO



amagi Flipkart 

Ashay Krishna
Director



 Microsoft

Harjeet Khanduja
Vice President



RELIANCE 

Shweta Shandilya
Director





Uday Narang
CEO



 OSM
OMEGA SEIKI MOBILITY

Jainendera Anand
CEO



 REVOLT

Rahil Gupta
CTO



Pushkaraj Salunke
CTO



Dheeraj Shetty
CHRO



Srikanth Reddy
CEO



Javed Khatri
CEO



Hyder Khan
CEO



Students Testimonials

The iScale assisted me in securing a placement within three months. The guidance and support from educators were incredible during my journey to become data scientist



Manas Jyoti Bohra



My learning experience has been great because, despite coming from a non-IT background, educators helped me understand the concepts and secure a Data Analyst job.



Shubham



Real-time, industry-oriented end-to-end projects are important and beneficial for me to mention on my resume.



Riya Garg



As a working professional, I easily manage my studies and job because the lectures are easy to understand, and the doubt assistance is really helpful for me



Suryakant



Transitioning from a non-IT BBA background to learning data analytics here is amazing because mock interviews and project sessions are very beneficial.



Anwar Khan



My learning experience is going well because I am gaining confidence through the interview preparation sessions, and the teaching method is incredible.



Abhinandan



I can learn anytime and anywhere because the lectures and study materials are easily accessible through the mobile app and web.



Vipin Verma



Here, educators skillfully integrate technical concepts with real-time, industry-oriented projects, enhancing our understanding through practical applications.



Bhoopendra



Students Testimonials

The iScale helped me land a job in data science with AI. Despite coming from a non-IT background, their excellent teaching style enabled me to learn, and secured a job.



Ashish



I wasted 4-5 lakh rupees on courses from other platforms, but taking the iScale course was my best decision. This course is the best among all the platforms.



Kartik



The industry-oriented tools that The iScale teaches are commendable. Other platforms don't cover them in as much depth. The iScale is an excellent platform



Micheal



I never did coding in my career, but when I joined The iScale premium course, coding became very easy for me. The teaching approach is fantastic.



Mahendra



The ease of explanation of concepts by educators at The iScale is truly incredible. The doubt support & projects have helped me excel in my career



Rukhsar



The instant doubt resolution at The iScale is impressive. The live doubt sessions greatly assist me in my learning process.



Sanjay



The way The iScale facilitates hands-on work on real-life end-to-end projects is incredibly helpful for developing practical skills



Shubham



The iScale is a trustworthy platform, and the curriculum and teaching methodology are very appropriate. They teach with honesty



Krunal



Students Testimonials

I'm from London, and The iScale offers excellent facilities for international learners with teaching better than many foreign platforms.



Abhineet



Thanks to The iScale's support, today I have secured a good job. After studying here, cracking interviews became much easier for me.



Aditya



I come from a commerce background, yet I learned so well at The iScale that I successfully landed a job, thanks to The iScale.



Amit



After joining the course from The iScale, I find the topics very easy to understand, and the teaching method is truly special.



Lakshami



The facilities and knowledge provided by The iScale are the best compared to other platforms, and I really like the content.



Mayurash



At The iScale, everything is taught in Hindi in a very simple way, and the courses are also affordable.



Gaurav



Whenever I have doubts, special sessions are conducted for me. This is the best part of The iScale, it provides personalized guidance.



Abkash



Coming from a non-technical background, I can still understand everything at The iScale thanks to their great teaching.



Mayur



Students Testimonials

I like the unique teaching style at The iScale because everything is taught here with great simplicity.



Amit



I am from a non-IT background, yet at The iScale I understand both concepts and coding very well.



Raj



I had taken Python lectures from other places before, but couldn't understand them. At The iScale, everything is so easy to understand



Swapnil



I liked The iScale's free course on YouTube, so I took the premium course and now understand everything even better. The iScale is the best.



Risabh



The iScale empowered me to learn despite my humble background, and I genuinely enjoy studying here.



Rohit



The end-to-end projects at The iScale teach you a lot in a practical way, and their teaching method is truly unique



Shiva



The iScale's excellent teaching method motivated me to learn Data Analytics in a detailed and thorough way.



Abhishek



The iScale's live doubt support is exceptional, making learning easy and effective.



Esha



Students Testimonials

Thanks to The iScale, I successfully transitioned my career and landed a data-driven job.



Abhinandan



At The iScale, the way of practicing questions and giving explanations is very simple and easy to understand.



Kishan



Whenever I have doubts, The iScale makes it easy to get them cleared through live classes.



Raja



The iScale's content and curriculum are very well-sequenced and organized, which makes learning interesting for me.



Rohit



At The iScale, all problems are solved during classes, which makes it easy for me to understand everything.



Mamta



The Educators at The iScale have great expertise in teaching, and I am very happy studying here.



Ritik



The iScale's content is aligned with industry standards, which makes learning relevant and effective



Vikram



The iScale not only teaches skills but also builds confidence, making every learner feel ready to take on real-world challenges.



Diksha



Talk With Community

We have conducted the Interview Experience Talk of the Achievers. Focused, hard work is the real key to success. Keep your eyes on the goal, and just keep taking the next step towards completing it.



TIGER
ANALYTICS

Ankush



Cognizant

Sharmine



CATERPILLAR

Siya



accenture

Manish



Deloitte.

Shreya



Itron

Kishan



accenturestrategy

Aman



ORACLE

Ashish



PayPal

Vrishank



MAERSK

Venus



IBM

Jahanvi



FiftyFive TECHNOLOGIES

Vishal

**Anand****Jashan****Ayushi****Roshan****Anindya****Vaibhav****Ankush****Nishant****Lokesh****Harshit****Pushpanjali**

There are approximately 400,000 bytes of data for every grain of sand on earth. For managing this huge data, Data Scientists , AI Professionals & Analysts' demands are growing globally.

CURRICULUM

Module 1

Fundamentals of Python

UNIT 1: The Basics of Python Programming

- Understanding Problem Solving
- Why Python
- What is an IDE (Integrated Development Environment)
- Installation of Python – Anaconda & Launching of Jupiter Notebook
- Write your first Python Program

UNIT 2: Variables and Operators in Python

- Python Arithmetic Operators
- Python Assignment Operators
- Python Boolean & Logical Operators
- Python Comparison Operators
- Python Bitwise Operator

UNIT 3: Built-in Functions of Python

- Round Function & Divmod Function
- Isinstance Function & Pow Function,
- Input Function , int function, Float Function
- Len, Min and Max function
- Sorted Function and help function

UNIT 4: Control Flow Statements

- If Condition
- If else Condition
- If-Elif-else Condition
- Nested if

UNIT 5: Concepts of Loops

- While Loop
- Break & Continue Statements
- For Loop & Range Function
- Nested Loops in Python

UNIT 6: User-Defined Functions in Python

- Introduction & Doc String
- Functions Input Arguments
- Functions Order of input Arguments
- Functions Return Statement
- Function Variable Number of Input Arguments
- *args and **kwargs in Function
- Lambda function

UNIT 7: Python Strings

- Introduction to String
- Indexing & Slicing
- Methods – strip, lower, upper
- Methods – replace, split, capitalize
- Methods – title, swapcase, find, index

- Methods – isalnum, isalpha, isdigit
- Methods – isupper, islower, istitle, startswith, endswith
- Multiline String & checking of substring in String
- Converting String into another Data Type
- Loops in string

UNIT 8: Python List

- Introduction to List
- Slicing on List
- Indexing on List
- Methods - Append, Extend & Insert in List
- Methods - Remove, Pop, Clear
- Methods – Count, Max,Min
- Reverse, Index Count, Copy
- Using Loops in List
- Using conditional statement in List

UNIT 9: Python Tuples

- Introduction, Concatenation & Nesting
- length, count, index method, indexing & slicing
- The tuple() Constructor
- Duplicates in tuples
- max, min, unpacking of tuples
- conditional statement and loops in tuple

UNIT 10: Python Sets

- Introduction to Set

- Methods - add, update, remove, discard, pop, clear, del, union
- Methods - intersection_update, intersection, difference_update, difference
- Methods - symmetric_difference_update, symmetric_difference
- Methods - isdisjoint, issuperset, issubset
- Methods - set(), max, min, copy, loops in sets

UNIT 11: Python Dictionary

- Introduction to Dictionary
- Methods-, len, dict constructor fromkeys, setdefault
- Methods - get, keys, values, items, update
- Methods - pop, popitem, del, clear, copy
- Methods - Nested Dictionaries, loops in dictionary

UNIT 12: Object Oriented Programming

- OOPS Basic concept & creating classes
- polymorphism and encapsulation
- Types of Inheritance - single, multilevel, multiple, hierarchical
- abstraction & decorator
- Types of methods - Class method, static method
- Special-magic/dunder method
- property decorators – getters, setters and deletes

UNIT 13: Files Handling

- Working with Files

- Reading & writing files
- Buffered Read and Write
- Other File Methods
- Logging & Debugger

UNIT 14: Exception Handling

- Introduction
- Exception handling with try, except, else, and finally block of code.
- Custom Exception Handling
- List of general use exceptions
- Best Practice Exception Handling

Module 2

Python Libraries

UNIT 1: NumPy

- Introduction to Numpy
- Numpy - Nd Array Object.
- Numpy - Data Types.
- Numpy - Array Attributes.
- Numpy - Array Creation Routines.
- Numpy - Array From Existing.
- Data Array From Numerical Ranges.
- Numpy - Indexing & Slicing.
- Numpy – Advanced Indexing.
- Numpy – Broadcasting.
- Numpy - Iterating Over Array.
- Numpy - Array Manipulation.
- Numpy - Binary Operators.
- Numpy - String Functions.
- Numpy - Mathematical Functions.
- Numpy - Arithmetic Operations.
- Numpy - Statistical Functions.
- Sort, Search & Counting Functions.
- Numpy - Byte Swapping.
- Numpy - Copies & Views.
- Numpy - Matrix Library.
- Numpy - Linear Algebra.

UNIT 2: Pandas

- Introduction to Pandas
- Reading Data From Different File Systems
- Python Pandas – Data Frame
- Python Pandas - operations
- Python Pandas - Basic Functionality
- Reading Data From Different File Systems
- Python Pandas – Re-Indexing Python
- Pandas – Iteration
- Python Pandas – Sorting.
- Working With Text Data Options & Customization
- Indexing & Selecting
- Data Statistical Functions
- Python Pandas - Window Functions
- Python Pandas - Date Functionality
- Python Pandas –Time Delta
- Python Pandas - Categorical Data
- Python Pandas – Visualization
- Python Pandas – Tools

UNIT 3: Matplotlib

- Introduction to Matplotlib
- Bar chart
- Scatter Plot
- Bar Graph
- Histogram
- Pie Chart
- Stem Plots
- Box Vs Whisker Plot
- Area Vs Stack Plot
- Step Plot
- Fill_Between Plotots

- subplot
- Savefig
- Axis matplotlib plots
- Text in matplotlib plots

UNIT 4: Seaborn

- Introduction to Seaborn
- Heatmap plot method
- Count plot method
- Violin Plot method
- Pair plot method
- Strip Plot method
- Box plot method
- Factor plot method
- Cat Plot method
- Styling plot method
- Facet Grid Method
- Line Plot method
- Histogram plot method
- Bar plot method
- Scatter plot method

UNIT 5: Plotly

- Introduction to Plotly
- Bubble Charts
- Bar charts
- Pie Charts
- Creating maps
- Time Series

Python Projects

Project 1: Google Image Scrapping - Scraping Google Images, involves extracting data from website the project involved sending requests to the Google Images website, parsing the HTML content, and extracting relevant image URLs using BeautifulSoup. This project showcases technical skills in Python programming and data analytics.

Project 2: Covid-19 Impacts Analysis - Covid-19 impacted the global economy. This project is to analyze the spread of Covid-19 cases and all the impacts of covid-19 on the economy.

Project 3: Movie-Recommendation System Project -

Recommendation systems are among the most popular applications of data science. This project is about the recommendation of movies.

Project 4: WhatsApp Chats Analysis - WhatsApp is one of the most used messenger applications today so we can use WhatsApp chats for analyzing our chat with a friend, customer, or a group of people.

Project 5: Smart Price Comparison Tool - In this project, you will develop a Python-based tool that compares product prices across different platforms or datasets. It will involve extracting and organizing data, applying comparison logic, and identifying the best available deals. You will also learn how to clean and process product information for accurate results. This project is highly useful in building skills for e-commerce, automation, and data handling.

Module 3

Statistics

UNIT 1:

- Introduction to Statistics
- Types of Statistics
- Types of Data
- Levels of Measurement

UNIT 2:

- Measures of Central Tendency
- Measures of Central Tendency in Python
- Measures of Dispersion
- Measures of Dispersion in Python
- Random Variables
- Sets
- Histogram Skewness
- Covariance Correlation
- Covariance Correlation in Python

UNIT 3:

- Probability Density Distribution Function
- PDF, PMF CDF
- Types of Probability Distribution
- Bernoulli Distribution

- Binomial Distribution
- Poisson Distribution
- Normal or Gaussian Distribution

UNIT 4:

- Uniform Distribution
- Z Stats Z Table
- Central Limit Theorem

UNIT 5:

- Estimate
- Hypothesis Testing Mechanism
- P value
- Z Test with Examples
- Student T Distribution
- T Stats T Test
- When to use T test vs Z test

UNIT 6:

- Type 1 Type 2 Error
- Confidence Interval Margin of Error
- Bayes Theorem

UNIT 7:

- What is Chi Square Test
- Chi Square for Goodness of Fit
- Chi Square Test with Python

UNIT 8:

- F Distribution
- Variance Ratio Test (F test)
- F test with Python

UNIT 9:

- Anova
- Assumptions in Anova
- Types of Anova
- Partition of variance in Anova

Module 4

Feature Engineering

UNIT 1:

- AI vs ML vs DL vs DS
- Supervised, Unsupervised Reinforcement Learning
- Train, Test Validation
- Variance, Bias, Overfitting Underfitting

UNIT 2:

- Handling Missing Values
- Handling Imbalanced Dataset
- SMOTE
- Data Interpolation
- Handling Outliers

UNIT 3:

- Feature Extraction
- Feature Scaling Normalization
- Normalization Min Max Scaling
- Unit Vectors Feature Scaling
- PCA

UNIT 4:

- Data Encoding
- Nominal or One Hot Encoding

UNIT 5:

- Label Ordinal Encoder
- Target Guided Ordinal Encoding

Module 5

Exploratory Data Analysis

UNIT 1:

- **EDA With Red Wine Data**

Exploratory Data Analysis (EDA) With Red Wine Data" is focusing on the analysis of a dataset related to red wine. Through statistical and visual techniques, this aims to uncover patterns, trends, and insights within the data.

- **EDA Student Performance Indicator**

EDA Student Performance Indicator involves conducting Exploratory Data Analysis (EDA) on a dataset related to student performance.

- **EDA Forest Fires**

EDA Forest Fires refers to an Exploratory Data Analysis project focusing on the examination of a dataset related to forest fires.

UNIT 2:

- **EDA Flight Price**

EDA Flight Price focuses on Exploratory Data Analysis for a dataset related to flight prices.

- **EDA Forest Fires**

EDA Google Playstore" involves conducting Exploratory Data Analysis on a dataset related to the Google Play Store.

Module 6

Machine Learning – I Supervised

Part 1 – Regression

UNIT 1:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Linear Regression

UNIT 2:

- R squared Adjusted R squared
- MSE, MAE RMSE
- Simple Linear Regression With Python
- Multiple Linear Regression

UNIT 3:

- Ridge Regression
- Lasso Regression

UNIT 4:

- Elastic Net Regression
- EDA with Algerian Forest Fire
- Model Training for Ridge, Lasso Elastic Net

Project 1: End to End ML Project :

Weather Prediction - Scraping Google Images, involves extracting data from website the project involved sending requests to the Google Images website, parsing the HTML content, and extracting relevant image URLs using BeautifulSoup. This project showcases

PART 2: Logistics Regression

- Logistic Regression Indepth Intuition
- Logistic Regression with Regularization
- Performance Metrics Confusion Matrix, Accuracy, Precision Recall
- Cross Validation Types
- Hyperparameter Tuning
- Logistic Regression Implementation
- Logistic Regression Multiclass Classification

Project 2: End to End ML Project :

Diabetes Prediction - Diabetes Prediction is a machine learning project for data science designed to forecast the likelihood of diabetes based on relevant health data. Using predictive modeling, the project aims to assist in early detection and proactive management of diabetes risk factors.technical skills in Python programming and data analytics.

PART 3: Decision Tree

- Decision Tree Classifier Intuition
- Decision Tree For Numerical Split

- Post Pruning And Prepruning Decision Trees
- Decision Tree Classifier Implementation
- Decision Tree Post Pruning
- Decision Tree Regressor In depth Intuition
- Decision Tree Regressor Implementation

PART 4: Support Vector Machines

- Support Vector Classifier Indepth Intuition
- Support Vector Machines Classifier
- Support Vector Regressor Indepth Intuition
- Support Vector Regressor Implementation
- SVM Kernels Intuition
- SVM Kernels Implementation

PART 5: Naive Bayes

- Naive Bayes Indepth Intuition
- Variants Of Naive Bayes Algorithms
- Naive Bayes Practical Implementation

Project 3: End to End ML Project :

Medical Prediction - • Medical Prediction employs machine learning for data science to forecast health outcomes based on relevant medical data. This project aims to enhance predictive analytics in the medical field for improved diagnosis and patient care technical skills in Python programming and data analytics.

PART 6: Ensemble Techniques & its types

- Ensemble Techniques And Bagging
- Random Forest Classifier And Regressor
- Out Of Bag Score Decision Trees
- Random Forest Practical Implementation

PART 7: Boosting

- Boosting Technique
- Gradient Bosting Indepth Intuition
- Xgboost Classification Algorithms
- Xgboost Regresor Algorithm

PART 8: KNN Algorithm

- KNN Classification And Regression
- Variants Of KNN
- KNN Classifier and Regr Implementation

Module 7

Machine Learning – II Unsupervised

PART 1: PCA

- Curse Of Dimensionality
- Geometric Intuition Behind PCA
- Mathematical Intuition Of PCA
- PCA Practical Implementation

PART 2: Clustering Algorithms

- K Means Indepth Intuition
- Hierarchical Clustering Intuition
- K means vs Hierarchical Clustering
- DBSCAN Clustering
- Silhouette Score Clustering
- K Means Clustering Implementation
- Hierarchical Clustering Implementation
- DBSCAN Algorithms Implementation

UNIT 3: Anomaly Detection

- Anomaly Detection Isolation Forests
- DBSCAN Anomaly Detection
- Local Outlier Factor Anomaly Detection

PART 4: Time Series

- Introduction of time series
- components of time series
- moving average
- stationary and non stationary
- acf pacf
- arima
- time series model building
- Time series EDA

Project 1: Diamond Price Prediction - Diamond Price Prediction is a machine learning project for data science that predicts diamond prices based on various features. Using algorithms, it aims to provide accurate and data-driven insights for pricing within the diamond industry.

Module 8

Deep Learning

PART 1: Neural Network & Its Perception

- Intro to Deep Learning Usecases
- Neural Network, Perceptron Mathematical Explanation

PART 2: Neural Network

- Mathematical Concepts
- Activation Functions
- Forward Back Propagation
- Implementation of ANN using Keras
- Callback Functions

UNIT 3: ANN

- Regression using ANN
- Loss Function
- Batch Normalisation
- Regularization

UNIT 4: Tensor Flow & Pytorch

- Regularisation in Deep Learning
- Weight Initialisation
- Optimizers

- Tensorflow
- Pytorch

Project 1: Credit card Fault detection :Credit Card Fault Detection is a deep learning project for data science focused on identifying fraudulent transactions within credit card data. Leveraging neural networks, the project aims to enhance the accuracy of fraud detection algorithms for secure financial transactions

PART 5: CNN

- Convolutional Neural Networks
- CNN Foundation 1
- CNN 2
- CNN 3 (Explainer)
- CNN 4 (LENET)
- CNN 5 (Alex Net)
- CNN 6
- CNN 7
- VGG NET
- Resnet
- Inception Net
- RCNN
- Fast RCNN
- Faster RCNN (Object Detection)
- Non Maximum Suppression

PART 6: YOLO

- YOLO
- YOLO V2
- YOLO V3
- YOLO V4
- YOLO V4 Part 2
- YOLO V5

PART 7: GAN

- GAN Introduction
- Training using GAN
- DC GAN
- Style GAN
- W Gan
- GAN Practical

Module 9

NLP

PART 1:

- Introduction to NLP
- History of NLP
- Why NLP
- Use of NLP
- web scrapping
- Text processing
- Understanding regex

PART 2:

- String Tokenization
- Sentence Processing
- Word Embedding
- Lemmatization in text processing
- Frequency Distribution
- Annotator creation
- Word Count
- Text Normalization
- Word 2 Vec
- Co Occurance Vector

PART 3: NLP

- Doc 2 Vec
- Text blob
- NLTK
- Genism
- RNN
- LSTM
- Bi LSTM

Project 1 – Smart Voice Response System - Speech-to-Speech Project, this project focuses on developing a system that takes spoken input and generates spoken output in another form, such as language translation or conversational responses. It highlights how speech technologies can be applied in communication tools, translation systems, and interactive assistants.

Project 2 – Voice-to-Text Converter - Speech-to-Text Project, this project focuses on building a system that converts spoken language into written text. It is commonly used in transcription services, voice-command applications, and digital assistants, making it an essential component in modern speech and language technologies.

Module 10

Computer Vision (Open CV)

Computer Vision

- **Computer Vision** in data science focuses on enabling machines to interpret and make decisions based on visual data. It involves developing algorithms and models to analyze images or videos, extracting meaningful information for applications like image segmentation, object detection, and scene recognition.
- **Detection of Objects** detection in computer vision is the process of identifying and locating specific objects within images or videos. Using advanced algorithms, this technology enables machines to recognize and draw bounding boxes around objects.
- **Scene Recognition** in computer vision involves teaching machines to understand and categorize the overall context or setting within an image. Through sophisticated algorithms, this technology enables systems to identify and classify scenes.
- **Image Segmentation** in computer vision is the process of dividing an image into meaningful segments or regions. This technique, facilitated by advanced algorithms, allows for precise identification and extraction of objects within an image.

Project 1: Object racking - Object Tracking in computer vision for data science involves developing algorithms to follow and analyze the movement of objects. This project aims to enhance real-time tracking capabilities for applications such as surveillance and autonomous systems.

Project 2: Image Classification - Image Classification" in computer vision for data science focuses on developing algorithms to categorize and label images based on their content. This project aims to improve automated image recognition systems for various applications.

Project 3: image to text - Image to Text in computer vision for data science involves converting visual content into text using machine learning algorithms. This project aims to enhance accessibility and information retrieval by enabling automated extraction of textual information from images

Module 11

Generative AI

• What is Generative AI

- Definition and Concept
- Difference between Traditional AI and Generative AI
- Examples of Generative AI in real life

• Key Applications in Data Science, NLP & Computer Vision

- Data Augmentation in Data Science
- Text Summarization and Translation (NLP)
- Image Generation and Recognition (Computer Vision)
- Chatbots and Virtual Assistants

• Generative Adversarial Networks (GAN)

- Generator and Discriminator Basics
- Training Process of GANs
- Applications: Image Synthesis, Deepfakes

• Transformers & Large Language Models

- Introduction to Transformers
- Self-Attention Mechanism
- Encoder-Decoder Structure
- Positional Encoding
- Role in Large Language Models (LLMs)

• Neural Networks

- Neurons and Layers
- Activation Functions
- Input, Hidden, and Output Layers
- Basic Working Principle

• CNN – Convolutional Neural Network

- Convolution Layer
- Pooling Layer
- Fully Connected Layer
- Applications in Image Processing

• RNN – Recurrent Neural Network

- Sequential Data Handling
- Hidden State Concept
- Limitations of RNNs (vanishing gradient)

• LSTM – Long Short Term Memory

- Cell State and Gates
- Forget Gate, Input Gate, Output Gate
- Solving Vanishing Gradient Problem
- Applications in Text & Speech

• Forward & Backpropagation

- Forward Pass Basics
- Loss Function Calculation
- Backpropagation and Weight Updates

• Text Generation

- Sequence-to-Sequence Models
- Language Modeling
- Applications: Chatbots, Content Generation

• TensorFlow

- Introduction and Features
- Tensors Basics
- Building Neural Networks in TensorFlow
- Applications in AI & ML

• PyTorch

- Introduction and Features
- Tensors in PyTorch
- Autograd Functionality
- Model Training with PyTorch

• Hugging Face

- Introduction to Hugging Face
- Transformers Library
- Pre-trained Models (BERT, GPT, etc.)
- Applications in NLP

• Transformers

- What are Transformers
- Self-Attention Basics
- Encoder and Decoder Blocks
- Applications in NLP and Vision

Project 1 – Text Transformation Project- In this project, you will build a model that transforms one piece of text into another, such as summarization or translation. It highlights the core power of Natural Language Processing (NLP) and strengthens your practical understanding of text processing. This project strengthens your practical understanding of Natural Language Processing (NLP) and helps you gain hands-on experience in text transformation tasks.

Project 2 – Text-to-Speech Conversion Project- This project focuses on converting written text into natural-sounding speech. It demonstrates the process of transforming text into voice and helps you gain hands-on experience in speech generation. It provides hands-on experience in creating voice-enabled systems and understanding the underlying concepts of text-to-speech technology.

Project 3 – Telegram Chatbot with LLM Integration- In this project, you will create a Telegram chatbot that connects with a Large Language Model (LLM). The chatbot will be able to answer questions and chat naturally, showing how AI can be used in everyday messaging apps.

Module 12

Agentic AI

• Agentic AI Introduction

- What is Agentic AI and how it differs from traditional AI systems
- Key characteristics: autonomy, adaptability, and intelligent decision-making
- Real-world examples of Agentic AI
- Benefits of Agentic AI in solving dynamic and complex problems
- Challenges and limitations in practical applications

• Agentic AI Lifecycle

- Stages of an AI agent
- Continuous learning and improvement in live environments
- Importance of performance evaluation and feedback loops
- Best practices for maintaining robust and scalable agentic systems

• Types of Agents

- Simple Reflex
- Model-Based Agents
- Goal-Based Agents
- Utility-Based Agents
- Comparison of agent types

• **Single vs. Multi-Agent Models**

- Definition and characteristics of single-agent systems
- Overview of multi-agent systems (MAS)
- Interaction patterns: cooperation, competition, and coordination
- Use cases of MAS in real-world applications
- Challenges in communication, conflict resolution, and scalability

• **Decision-Making Mechanism of Agentic AI**

- How agents perceive their environment using sensors or data inputs
- Reasoning and planning mechanisms to select optimal actions
- Execution of actions to achieve objectives
- Handling uncertainty and dynamic environments
- Feedback loops and learning from outcomes

• **Large Language Models (LLMs)**

- Role of LLMs as the “brain” of agentic AI
- Understanding natural language and generating human-like responses
- Planning and reasoning tasks using LLMs
- Examples of LLM-driven agents in practice
- Limitations of LLMs and strategies to improve reliability

• Retrieval-Augmented Generation (RAG)

- Concept of RAG and why it enhances LLM performance
- How RAG combines retrieval and generative models
- Applications in question-answering, customer support, and research
- Benefits and challenges of implementing RAG

• Text Embedding

- What is text embedding and why it's important?
- How embedding convert text into numerical vectors for machines
- Role in semantic search, similarity detection, and information retrieval
- Examples of embedding models and libraries
- Best practices for creating high-quality embedding

RAG Models

- Architecture of RAG: retrieval system + generative model
- How RAG retrieves relevant information and generates coherent answers
- Use cases in enterprise AI, research, and knowledge management
- Integration with agentic AI workflows
- Advantages over traditional LLM-only approaches

• **LangChain Framework**

- Introduction to LangChain for building agentic applications
- Core components: chains, agents, and memory
- How it simplifies integration of LLMs with external tools
- Hands-on examples of building workflows with LangChain
- Benefits of using LangChain in production environments

• **OLLAMA**

- Running large language models locally with OLLAMA
- Benefits: privacy, speed, offline access, and customization
- Supported LLMs and deployment options
- Use cases for research, experimentation, and enterprise applications
- Best practices for local model optimization

• **Langflow Framework**

- Visual, drag-and-drop interface for building AI workflows
- Rapid prototyping without extensive coding knowledge
- Integrating LLMs, APIs, and data sources into agentic pipelines
- Experimentation, testing, and deployment of workflows
- Advantages of Langflow for productivity and collaboration

• DataStax

- Introduction to DataStax as scalable vector databases
- Importance for storing and querying large amounts of AI data
- Integration with LLMs and RAG pipelines
- Enterprise-grade applications for AI solutions
- Best practices for database design, indexing, and performance

Project 1 – RAG AI Agent (Retrieval-Augmented Generation Agent)

Learn how to build an AI agent that combines information retrieval with text generation. This project will show how AI can fetch relevant data from documents or knowledge bases and generate accurate, contextual responses – just like modern AI chatbots.

Project 2 – Plant Disease Detection AI Agent- Work on a real-world computer vision project where AI identifies plant diseases from leaf images. This project is highly practical in agriculture, showing how AI can support farmers with early disease detection and better crop management.

Project 3 – Personal AI Assistant Agent- Build your own AI-powered virtual assistant that can answer queries, manage tasks, and interact naturally with users. This project demonstrates how AI can enhance productivity and provide personalized support in daily life.

Module 13

20+ AI Tools

- **Introduction to Prompt Engineering**

Introduction to Prompt Engineering covers the fundamentals of designing effective prompts to interact with AI models. It includes structuring questions, providing context, and experimenting with different prompt techniques to generate accurate and relevant AI outputs.

- **ChatGPT**

A versatile AI assistant by OpenAI that excels in answering questions, writing content, coding, and problem-solving. It is widely used for learning, automation, and boosting productivity.

- **Gemini**

Google's powerful multimodal AI model that can understand text, images, and code. It is best for research, reasoning, and creating advanced AI-driven applications.

- **NotebookLM**

An AI research assistant that summarizes, organizes, and explains documents. It is highly useful for students, researchers, and professionals to study and work efficiently.

• Grok

Known for its lightning-fast AI processing speed, making responses almost real-time. It is mainly used for chatbot's, analytics, and high-performance AI tasks.

• Perplexity

An AI-powered search engine that delivers accurate answers with cited sources. It is ideal for research, fact-checking, and exploring topics in depth.

• Claude

Anthropic's AI assistant designed with safety and reasoning at its core. It is used for thoughtful writing, coding help, and intelligent conversations.

• Gamma AI

A tool that creates professional presentations, documents, and webpages quickly. It is best for professionals and students who need high-quality designs effortlessly.

• MidJourney

An AI image generator that creates stunning and artistic visuals from text prompts. It is widely used by designers, marketers, and creative professionals.

• Leonardo AI

A creative design tool for generating game assets, digital art, and graphics. It is ideal for artists and content creators looking for customizable visuals.

• Freepik

A platform offering AI-generated stock photos, vectors, and illustrations. It is useful for marketers, designers, and businesses that need ready-to-use visuals.

• Krea AI

A creative AI tool that generates unique images, animations, and concept art. It is best for storytellers, designers, and digital creators.

• OpenAI Sora

A groundbreaking AI that transforms text into realistic, high-quality videos. It is perfect for filmmakers, advertisers, and storytellers.

• Google AI Studio

A developer platform to test, explore, and deploy Google's AI models. It is mainly used for building, scaling, and integrating AI applications.

• Notion AI

Enhances the Notion workspace by helping with writing, summarizing, and brainstorming. It is highly useful for productivity, project management, and note-taking.

- **Nano Banana**

A lightweight AI designed for fast and efficient image generation. It is best for quick creative experiments and visual content creation.

- **Manus AI**

An intelligent writing assistant that improves grammar, tone, and clarity. It is widely used for professional emails, reports, blogs, and academic writing.

- **DeepSeek**

An open-source AI platform offering efficient and reliable models. It is best for coding, data analysis, and advanced research tasks.

- **Meta AI**

Provides innovative open-source models like LLaMA for AI development. It is mainly used by researchers and developers for custom AI solutions.

- **Ollama**

A platform that allows users to run AI models locally on their devices. It is great for privacy-focused users who want customizable AI.

- **QuilBot**

A writing tool for paraphrasing, summarizing, and grammar correction, perfect for students, writers, and professionals.

Module 14

SQL

UNIT 1: Introduction and Installation Software

- Roadmap to learn SQL & syllabus discussion
- Download & Install MySQL
- Download & Install XAMPP
- Download & Install PostgreSQL & Pgadmin

UNIT 2: Fundamentals of SQL

- Introduction to SQL
- Database Fundamental
- Learning Resources - (Cheat Code | Practice Dataset | PDF Notes & Books | Learning Website | Interview Questions)

UNIT 3: Case Study with Example

- Create Database | Drop Database | Create Table | Insert Table | Drop Table
- Case study with XAMPP Software
- Case study with MySQL software
- Case study with PostgreSQL

UNIT 4: Data Types and Keys

- All Data Type in SQLs)
- ER / Database Diagram Example: Amazon Ecommerce
- Database Keys
- Cardinality of Relationships

UNIT 5: DDL Commands for SQL

- DDL Commands for Databases: Create | Drop
- DDL Commands for Table: Create | Truncate | Drop
- Data Indignity

UNIT 6: Constraints

- Constraints in MySQL (NOT NULL | UNIQUE | PRIMARY KEY)
- Constraints in MySQL (AUTO INCRIMENT | CHECK | DEFAULT | FOREIGN KEY)
- Referential Actions (Restrict | CASCADE | SET NULL | SET DEFAULT

UNIT 7: Alter Table

- ADD Columns in Alter Table
- DELETE Columns in Alter Table
- MODIFY Columns in Alter Table

UNIT 8: ADD, Editing and Deleting Constraints

- ADD Constraints
- DELETE Constraints
- EDIT Problems in MySQL, Drop Constrains

UNIT 9: INSERT Query

- INSERT query
- INSERT query variation
- INSERT multiple values

UNIT 10: SELECT Query

- SELECT all columns
- Filter columns
- Alias | renaming columns
- Create expression using columns
- Constant value
- DISTINCT(unique) values from a column
- DISTINCT Combinations
- Filter rows WHERE clause
- BETWEEN operator
- IN and NOT IN Operator

UNIT 11: UPDATE Query

- Update Query to update row(s)
- Update Multiple Columns

UNIT 12: DELETE

- DELETE Query to delete row(s)
- Deletion based on multiple conditions

UNIT 13: Aggregate Functions

- MAX() & MIN()
- AVG()
- SUM()

UNIT 14: Scalar Functions

- ABS()
- ROUND()
- CEIL() & FLOOR()

UNIT 15: SQL Joins

- What are SQL joins?
- Types of Joins
- Cross Joins --- Cartesian Products
- Inner Joins
- Right Joins
- Left Joins
- Full Outer Join
- Set Operations
- Self-Join
- Filtering Columns
- Filtering Rows

UNIT 16: Subquery

- What is Subquery?
- Scalar Subquery
- Row Subquery
- Table Subquery
- Correlated Subquery

UNIT 17: Sorting

- Fundamental of SQL Sorting
- Sorting top 10 data
- Sorting Descending Orders
- Sorting functions in datasets

UNIT 18: Grouping

- Fundamental of SQL Grouping
- Grouping Average Function
- Group By on multiple columns
- Sorting Descending Orders
- Sorting functions in datasets

UNIT 19: Project & Case Study:

Project 1: Indian Premier League (IPL Dataset)

- Case Study in Indian Premier League
- Find the top 5 batsmen in IPL
- Find the 2nd highest 6 hitters in IPL
- Find Batsman (M.S Dhoni, Virat Kohli) performance against all IPL team
- Find the top 10 batsmen with centuries in IPL using Sub-Quires
- Find the top 5 batsmen with the highest strike rate

UNIT 20: Project & Case Study:

Project 2: Shipping E-Commerce (Sorting: Smart Phone Datasets)

- Find the top 5 Samsung phones with the biggest screen size
- Sort all the phones in descending order of the number of total cameras
- Sort data on the basis of ppi in decreasing order
- Find the phone with 2nd largest battery
- Find the name and rating of the worst rated apple phone
- Sort phones alphabetically and then on the basis of rating in desc order
- Sort phones alphabetically and then on the basis of price in ascending order

Project 3 – Music Store Analysis - In this project, you will work with a music store database to practice SQL queries. It includes tasks like retrieving customer details, analyzing sales records, and tracking popular tracks and artists. You will also learn how to apply joins, aggregate functions, and filtering to extract meaningful insights. This project strengthens SQL skills through real-world data analysis.

Module 15

Power BI

UNIT 1: Introduction to Power BI

- Complete Overview of Power BI
- The Power of Data Visualization
- What are the BI & BI tools?
- What is Power BI
- Why Power BI? The Modern BI Choice
- Power BI: Key Advantages

UNIT 2: Creating Charts

- Introduction to Basic Charts
- Create a Column Chart
- Create a Stacked Column Chart
- Create a Pie Chart
- Create a Donut Chart
- Create a Ribbon Chart
- Create a Funnel Chart
- What are the Include and Exclude
- View selected Data and Export in CSV
- Overview

UNIT 3: Tables and Matrix

- Introduction to Table and Matrix
- How to create a Table & Formatting and Styling of a Tables
- How to use Conditional Formatting in Table
- How to change Aggregations
- How to Create a Matrix & Formatting and Styling of a Matrix
- How to use Conditional Formatting in Matrix
- How to create Hierarchies in Matrix
- Add Total and Sub-Total in Matrix
- Key Difference Between Matrix & Tables
- How to change Number Formatting

UNIT 4: Cards and Filters

- What is Cards and Filters in Power BI
- How to create a Number Card
- How to create a Date Card
- How to create a Text Card
- How to create a Multi-Row Card
- How to Format a Card
- How to apply Filter on Visual
- How to apply Filter on this Page
- How to apply Filter on All Pages
- How to use Drill Through

UNIT 5: Slicers

- Introduction to Slicers and its types
- How to create and format Text Slicers
- How to create Date Slicers
- How to format Date Slicers
- Creating a Number of Slicers

UNIT 6: Advance Charts

- Introduction to Advance Charts in Power BI
- How to create a Waterfall Chart
- How to create a Guage Chart
- How to create a Line Chart
- How to create a Stacked Area Chart
- Line vs Stacked Column Chart
- Line vs Clustered Column Chart
- What are Plots and How to create a Scattered Plot
- How to create a Tree Map
- Line Chart drill with Slicers

UNIT 7: Maps

- Introduction to Maps in Power BI
- How to create a Basic Map
- How to create a Filled Map
- How to create a Map with pie chart
- How to format a Map

UNIT 8: KPI & Other Functions

- Introduction to Advance Function
- How to use KPI Visualization Tool
- Learn Updates in MS Power BI
- Create a Key influencer in Power BI
- Create a Decomposition Tree in Power BI
- Create a Python & R Scripting in Power BI
- Third-Party Extensions 3 Icons Introduction in Power BI

UNIT 9: Insert & Action Functions

- Introduction Insert & Action
- How to Insert Image in Power BI
- Create Insert Text in Power BI
- How to Insert Shapes in Power BI
- Create Insert Buttons in Power BI
- Use Web URL Action in Power BI
- Work on Page Navigation Action in Power BI
- Use Bookmark Action in Power BI
- How to create Drill through Action in Power BI
- Create Power BI Account

UNIT 10: Advanced Extra Charts

- Introduction to Advanced Extra Charts in Power BI Desktop
- Create Scroller in Power BI
- Create a Word Cloud in Power BI

- Create Infographics in Power BI
- Create Drill-Down Donut Chart in Power BI
- Create Drill-Waterfall Chart in Power BI
- Create Play Axis Slicer in Power BI
- Create an Animated Bar Chart Race in Power BI
- Create Sunburst Chart in Power BI
- Create Sankey Chart in Power BI
- Create Calendar & Timeline Slicers

UNIT 11: Create a Superstore Report

- Create reports and dashboards documentation.
- Collaborate, share and integrate across products documentation.
- Get Power BI samples.
- Use report themes.
- Add conditional table formatting.
- Ways to share your work.
- Organize work in the new workspaces.
- Publish to web.

UNIT 12: Dashboard Functions

- How to Publish Report to Power BI Service
- Export Power BI Report into PPT, PDF & PBIX format
- Create a Pin Live Dashboard in Power BI
- What is Comment, Share & Subscribe in Power BI Reports
- How to Refresh Reports Automatically in Power BI Service
- Free Data Files Source for Practice

Power BI Projects

Project 1: Real Estate Data Exploration- This project focuses on analyzing real estate data using Power BI to uncover trends, patterns, and insights. Learners will create interactive dashboards to visualize property prices, demand, and regional comparisons. The project highlights how data-driven insights can support smarter investment and business decisions in the real estate industry.

Project 2: IPL Data Analytics- This project uses Power BI to analyze IPL data and uncover insights into team performance, player statistics, and match trends. Interactive dashboards help visualize patterns, compare results, and support data-driven decisions in the world of cricket.

Project 3: Analysis of Amazon Global Superstore- This project focuses on analyzing the Amazon Global Superstore dataset using Power BI to uncover sales trends, customer behavior, and product performance. Interactive dashboards visualize key metrics, enabling data-driven insights for better business decisions.

Project 4: OTT Analytics and Insights- This project uses Power BI to analyze OTT platform data, uncovering viewer behavior, content performance, and engagement trends. Interactive dashboards help visualize insights, enabling data-driven strategies for content optimization and audience growth.

Project 5 - UNICEF Global Data Analysis: Electricity Access and Poverty-

- This project uses UNICEF's global data on electricity access and poverty, analyzed through Power BI. It involves building an interactive map to visualize poverty and electricity access worldwide, and studying the correlation between access rates and poverty levels to highlight regions most in need of support.

Module 16

Advance Excel

UNIT 1: Fundamental of Excel: Beginner Level

- Basic Features - Introduction
- Basic Formatting
- Using Formulas
- Save File
- Filter & Sorting
- Conditional Formatting
- Insert & Delete Columns or Rows
- Find & Remove Duplicates
- Merge and Centre

UNIT 2: Intermediate Level of Excel

- Rounding of Numbers
- Autofill in Excel
- Add or Edit Comment
- Filters for Data Manipulation
- Sorting on multiple columns
- Insert Table
- Slicers

UNIT 3: Moderate Level of Excel

- Creating an Excel Column Chart
- Working with Excel Pie Charts
- Chart Group (Line, Area, Waterfall)
- Insert Picture or Shapes
- Add link or hyperlink
- Excel Sparkline's
- Add Drop Down List
- Split Text
- Page Layout Tab and Ribbon
- Printing Excel Worksheet

UNIT 4: Advance Level of Excel

- Formula Tabs & Library
- Define Names
- Formula Auditing
- Calculation Options & Watch Window
- Get and Transform Data
- Queries and Connections
- Excel data validation
- Excel PivotTables
- Excel conditional functions
- Excel Text based Functions
- Excel "What if?" Tools
- Conditional IF Function IN [EXCEL]
- Result Format with Nested-IF [Excel]
- XOR Formula (Exclusive OR) EXCEL

- Excel VLOOKUP Function
- Advance VLOOKUP Formula

UNIT 5: Data Processing and Create Visualization Dashboard

Project Title: Analysis the Online Store Annual Report Like Mantra | AJio | Amazon | Etc.

Description: Importing data from text files Create a fully functional online store dashboard with the analysis the store annual reports and learn data cleaning, processing, analysis and dashboard creation in excel.

Module 17

Tableau

Tableau is a business intelligence tool which is widely used in the analytics industry. It is one of the top data visualization tools used for data analysis and getting data driven business insights. Learn various techniques to extract meaning from large datasets, detailed approaches to design dashboards. Learn to create storyboards using Tableau which enables us to transform complex data into meaningful insights.

UNIT 1: Creating Visualizations

- Text Table and Formatting Visuals
- Heatmap and Highlight table
- Lines Area Chart
- Circle, Side by Side Circle View and Shapes
- WordCloud
- Scatter Plot
- Histogram Bins
- Gantt view and Waterfall Chart
- Reference lines and bands, Distribution band Bullet graphs
- Dual Axes Visuals Part 1
- Dual Axes Part 2

UNIT 2: Filtering, Sorting & Connecting

- Animated Charts

- Funnel Charts its types
- Donut Chart Multiple Donut Chart
- Dumbbell, Barbell or DNA Chart
- Butterfly Chart
- Box Whisker Plot
- Filters in tableau part 1
- Filters in tableau part 2

UNIT 3: Dashboard & Actions

- Parameters in tableau
- Sets in tableau
- Table Calculation
- Joins & Unions
- Data Blending
- Tableau dashboard
- Tableau Server
- User Security

UNIT 4: Review of Tableau Desktop

- Bar chart
- Line chart
- Scatter plot
- Geographic map
- Filled map
- Crosstab report
- Side by side bar

- Combines axis
- Stacked bar
- Dual axis
- Bar in bar
- Heat map
- Highlight table
- Pie chart
- Tree map
- Motion chart
- Histogram
- Dashboard
- Groups
- Sets
- Dates
- Calculations
- Quick table
- Parameters
- Reference lines

Tableau Projects

Project 1: FIFA World Cup Trends - This project leverages Tableau to analyze FIFA World Cup data, exploring team performance, player statistics, and tournament trends. Interactive visualizations provide clear insights, helping understand patterns and make data-driven observations about the games.

Project 2: Analytics for Sports Performance - This project uses Tableau to analyze sports data, focusing on player performance, team statistics, and game trends. Interactive dashboards help visualize insights, enabling data-driven decisions to enhance performance and strategy in sports.

Project 3: London Automobile Ride Analysis - This project uses Tableau to analyze automobile data, examining vehicle performance, sales trends, and market patterns. Interactive dashboards help visualize insights, enabling data-driven decisions for improving product strategy and business performance.

Module 18

Placement Preparation

UNIT 1:

- **Crafting an impressive Resume -**

Crafting an ATS-Friendly Resume - A resume is a summary of your career, skills, and achievements. In this program, you'll also learn how to create a resume that is ATS (Applicant Tracking System) friendly. An ATS is the software most companies use to filter and organize resumes, so making your resume ATS-friendly ensures it is easily readable by these systems as well as by recruiters.

- **Portfolio Building -**

A portfolio helps you to organize information so that you're always ready to apply for jobs. Your portfolio creates an image of you in an employer's mind. In this curriculum, you will learn about building the best portfolio.

- **Interview Tips -**

Interview tips will help you gain valuable insights into effective communication and key strategies to showcase your skills confidently.

Online face-to-face mock interviews help you train & prepare for job interviews in a no-pressure, stress-free environment simulating a real job interview. Mock interviews will be conducted for you in this course.

Subjects	Master of Data Analytics Program	Full Stack Data Science with Generative AI Program
1. Fundamental of Python	✓	✓
2. Advance Python	X	✓
3. Basic Python Libraries	✓	✓
4. Advance Python Libraries	X	✓
5. Statistics	X	✓
6. Feature Engineering	X	✓
7. Exploratory Data Analysis	✓	✓
8. Machine Learning - & Supervised + Unsupervised	X	✓
9. Deep Learning	X	✓
10. Natural Language Processing – NLP	X	✓
11. Computer Vision (Open CV)	X	✓
12. Generative AI + Agentic AI	X	✓
13. 20+ AI Tools	✓	✓
14. SQL	✓	✓
15. Power BI	✓	✓
16. Advanced Excel	✓	✓
17. Tableau	✓	✓
18. Placement Preparation Module	✓	✓

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